

Integrating LLMs into Clinical Reasoning

Workshop on 16 June 2026

1. Concept

Clinical reasoning involves interpreting symptoms and transforming them into coherent diagnostic and therapeutic decisions.

Following Daniel Kahnemann, cognitive processing in humans can be described through two distinct pathways – intuitive, unconscious and emotional (System 1) and deliberate, conscious and analytical (System 2). Modern Large Language Models (LLM) excel in System 1 “thinking” through fast, automatic decision making by virtue of learning associations from billions of texts used in their training. However, they are not yet strong in System 2 symbolic thinking, which should follow rules (based on logical and/or Bayesian inference) to evaluate options and evidence, as this is critical for a more accurate and reflective decision making. A related challenge concerns cases where LLMs make right decisions for wrong reasons. As LLMs lack direct connections to the intentions and values underlying human communication, they often take System 1 shortcuts based on specific keywords due to distributional anomalies in their training datasets.

This concept will result in a novel approach to make clinical LLMs more reliable by explicitly integrating mechanisms for symbolic, explainable, and causally grounded System 2 clinical reasoning rather than relying solely on associative performance.

2. Plan

A one-day workshop will be a meeting point between healthcare professionals with expertise in clinical reasoning and AI researchers involved in designing LLM applications.

We will plan two one-hour keynote talks (the names to be confirmed), breakout sessions and a summary-reporting session. The breakout sessions will mix the delegates with clinical expertise and those with the AI background.

The task for the breakout sessions is to discuss how medical narratives about patients and their conditions translate into reasoning formalisms, and under which circumstances LLMs can help or hinder this translation. The resulting neuro-symbolic models will help understand causal drivers to prevent models from over-weighting spurious correlations. We expect that explicit symbolic rules or Bayesian inference rules can act as causal anchors, ensuring that the decisions reflect recognised clinical constructs (e.g., DSM-based criteria, BNF clinical guidelines or medical test interpretations) rather than hallucinations resulting

from associations. We can also think about utilising available large-scale formal ontologies such as SNOMed or OBO Foundry.

3. Outputs

This hands-on workshop should lead to establishing new connections with the aim of enabling ongoing collaboration, starting new funded projects, and, in the short term, producing research papers from the experience of interaction in the breakout rooms.

Organisers

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